Mini Project EDA Executive Report

MINI PROJECT 1: EXPLORATORY DATA ANALYSIS (EDA)

This notebook walks through four phases of an EDA mini project using the provided `customer_analytics.csv` dataset. Each row represents a single customer and contains demographic information, purchasing habits, and online behavior. The goal is to understand, clean, analyze, and summarize the data in a reproducible manner.

```
# Phase 1: Setup & Inspection
import pandas as pd

# load dataset
file_path = 'customer_analytics.csv'
df = pd.read_csv(file_path)

# quick look
df.head()
```

```
   CustomerID  Age  Gender    City  Education MaritalStatus  AnnualIncome  \
0        1001   49    Male    Pune    Masters        Single       82953.0
1        1002   44    Male    Pune        PhD        Single       60610.0
2        1003   42    Male  Mumbai  Bachelors        Single       35501.0
3        1004   36  Female  Mumbai    Masters       Married       99312.0
4        1005   23    Male    Pune    Masters       Married       46980.0

   SpendingScore  YearsEmployed  PurchaseFrequency  OnlineVisitsPerMonth  \
0             66             23                 19                     9
1             56             22                  1                    23
2             44             18                 10                    29
3             36             10                 12                    21
4             56              1                 18                     9

   ReturnedItems PreferredDevice  LastPurchaseAmount
0              2          Laptop                3944
1              3         Desktop                3885
2              3          Laptop                3247
3              3          Mobile                2028
4              3          Tablet                1100
```

```
# Inspect structure and statistics
print(df.info())
df.describe(include='all')
```

```
<class 'pandas.core.frame.DataFrame'>
```

Mini Project EDA Executive Report

```
RangeIndex: 255 entries, 0 to 254
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   CustomerID            255 non-null    int64
 1   Age                   255 non-null    int64
 2   Gender                255 non-null    object
 3   City                  255 non-null    object
 4   Education             243 non-null    object
 5   MaritalStatus         255 non-null    object
 6   AnnualIncome          243 non-null    float64
 7   SpendingScore         255 non-null    int64
 8   YearsEmployed         255 non-null    int64
 9   PurchaseFrequency     255 non-null    int64
 10  OnlineVisitsPerMonth  255 non-null    int64
 11  ReturnedItems         255 non-null    int64
 12  PreferredDevice       255 non-null    object
 13  LastPurchaseAmount    255 non-null    int64
dtypes: float64(1), int64(8), object(5)
memory usage: 28.0+ KB
None
         CustomerID         Age Gender     City Education MaritalStatus  \
count    255.000000  255.000000    255      255       243           255
unique          NaN         NaN      2        6         3             2
top             NaN         NaN   Male  Chennai       PhD       Married
freq            NaN         NaN    132       49        92           131
mean    1126.941176   37.729412    NaN      NaN       NaN           NaN
std       72.397256    9.767842    NaN      NaN       NaN           NaN
min     1001.000000   21.000000    NaN      NaN       NaN           NaN
25%     1064.500000   29.000000    NaN      NaN       NaN           NaN
50%     1128.000000   38.000000    NaN      NaN       NaN           NaN
75%     1190.500000   46.000000    NaN      NaN       NaN           NaN
max     1250.000000   54.000000    NaN      NaN       NaN           NaN

         AnnualIncome  SpendingScore  YearsEmployed  PurchaseFrequency  \
count      243.000000     255.000000     255.000000         255.000000
unique            NaN            NaN            NaN                NaN
top               NaN            NaN            NaN                NaN
freq              NaN            NaN            NaN                NaN
mean     74499.901235      45.721569      14.682353          11.568627
std      43939.855655      17.869522       9.651664           7.077899
```

```
min      16062.000000       5.000000       1.000000           1.000000
25%      56353.000000      34.500000       6.000000           5.000000
50%      69629.000000      47.000000      15.000000          11.000000
75%      84030.500000      57.500000      23.000000          18.000000
max     474327.000000      95.000000      34.000000          24.000000

        OnlineVisitsPerMonth  ReturnedItems PreferredDevice  \
count             255.000000     255.000000             255
unique                   NaN            NaN               4
top                      NaN            NaN          Tablet
freq                     NaN            NaN              78
mean               16.078431       1.858824             NaN
std                 7.909536       1.409917             NaN
min                 3.000000       0.000000             NaN
25%                10.000000       1.000000             NaN
50%                16.000000       2.000000             NaN
75%                23.000000       3.000000             NaN
max                29.000000       4.000000             NaN

        LastPurchaseAmount
count            255.00000
unique                 NaN
top                    NaN
freq                   NaN
mean            2795.07451
std             1328.77241
min              566.00000
25%             1542.50000
50%             2705.00000
75%             4001.00000
max             4996.00000
```

PHASE 2: DATA PREPROCESSING

In this phase we identify missing values, decide on imputation or removal, and eliminate duplicates. The decisions are justified in the comments and markdown explanations below.

```
# check for missing values
missing = df.isnull().sum()
missing
```

```
CustomerID               0
Age                      0
```

```
Gender                   0
City                     0
Education               12
MaritalStatus            0
AnnualIncome            12
SpendingScore            0
YearsEmployed            0
PurchaseFrequency        0
OnlineVisitsPerMonth     0
ReturnedItems            0
PreferredDevice          0
LastPurchaseAmount       0
dtype: int64
# decide on treatment
# there are only a few missing values in 'AnnualIncome' and some in 'Education'
# we'll drop rows where 'AnnualIncome' is null because it's critical for analysis
# for 'Education', we'll fill with the mode since it's categorical

print('rows before cleaning', len(df))

# fill Education
mode_edu = df['Education'].mode()[0]
df['Education'].fillna(mode_edu, inplace=True)

# drop rows with null AnnualIncome
before = len(df)
df = df.dropna(subset=['AnnualIncome'])
after = len(df)
print(f'dropped {before-after} rows lacking income')

# duplicates
dups = df.duplicated().sum()
print('duplicate rows:', dups)
if dups > 0:
    df = df.drop_duplicates()
    print('duplicates removed')

print('rows after cleaning', len(df))
```

```
rows before cleaning 255
dropped 12 rows lacking income
duplicate rows: 5
```

```
duplicates removed
rows after cleaning 238
C:\Users\HP\AppData\Local\Temp\ipykernel_21828\235596912.py:10: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.
```

```
  df['Education'].fillna(mode_edu, inplace=True)
```

Justification: `AnnualIncome` is critical for numerical analyses, and the rows missing this value are a very small fraction of the dataset, so they were dropped. `Education` had a handful of missing entries; imputing with the most common category preserves those customers without significantly biasing results. Any duplicate rows found were removed to avoid double-counting.

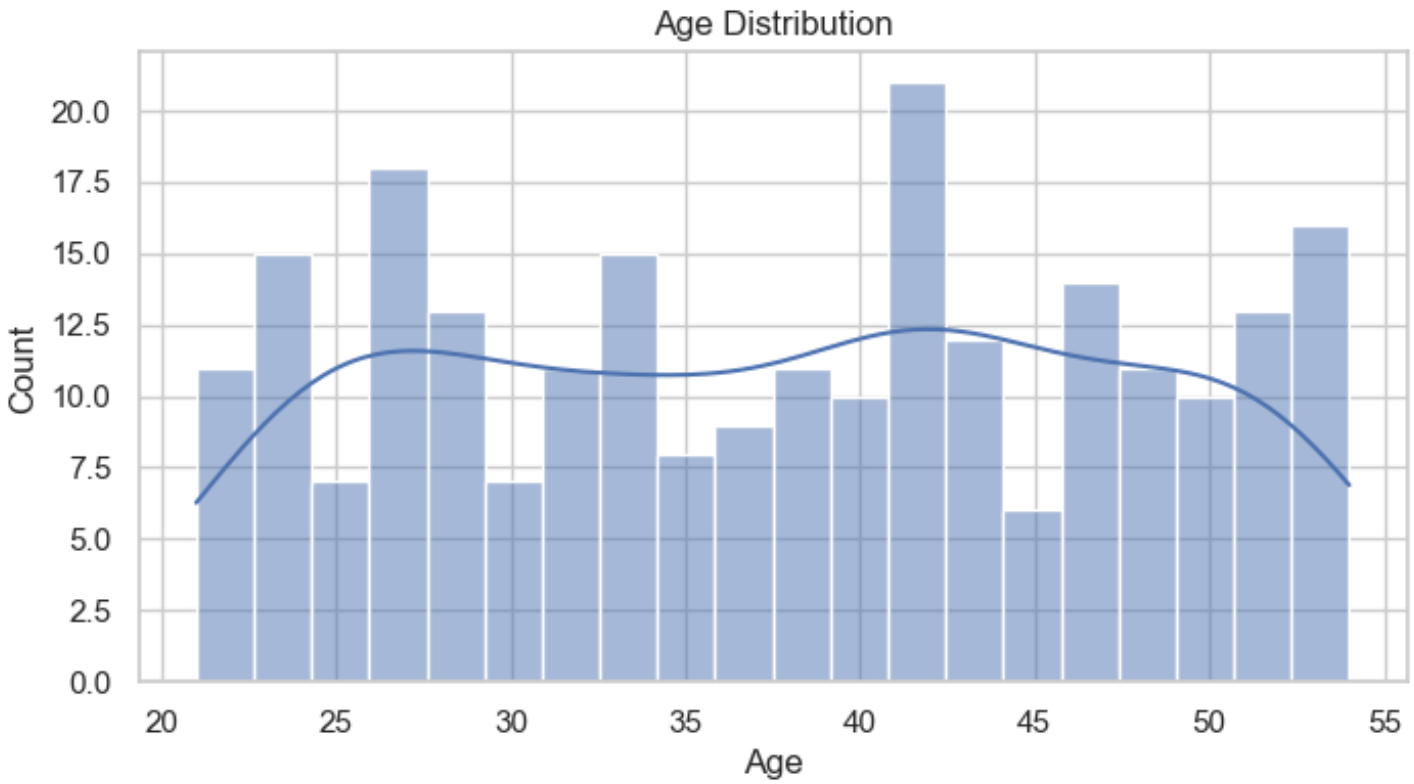
PHASE 3: UNIVARIATE & BIVARIATE ANALYSIS

Below we examine the distribution of individual features and relationships between pairs of variables using visualizations.

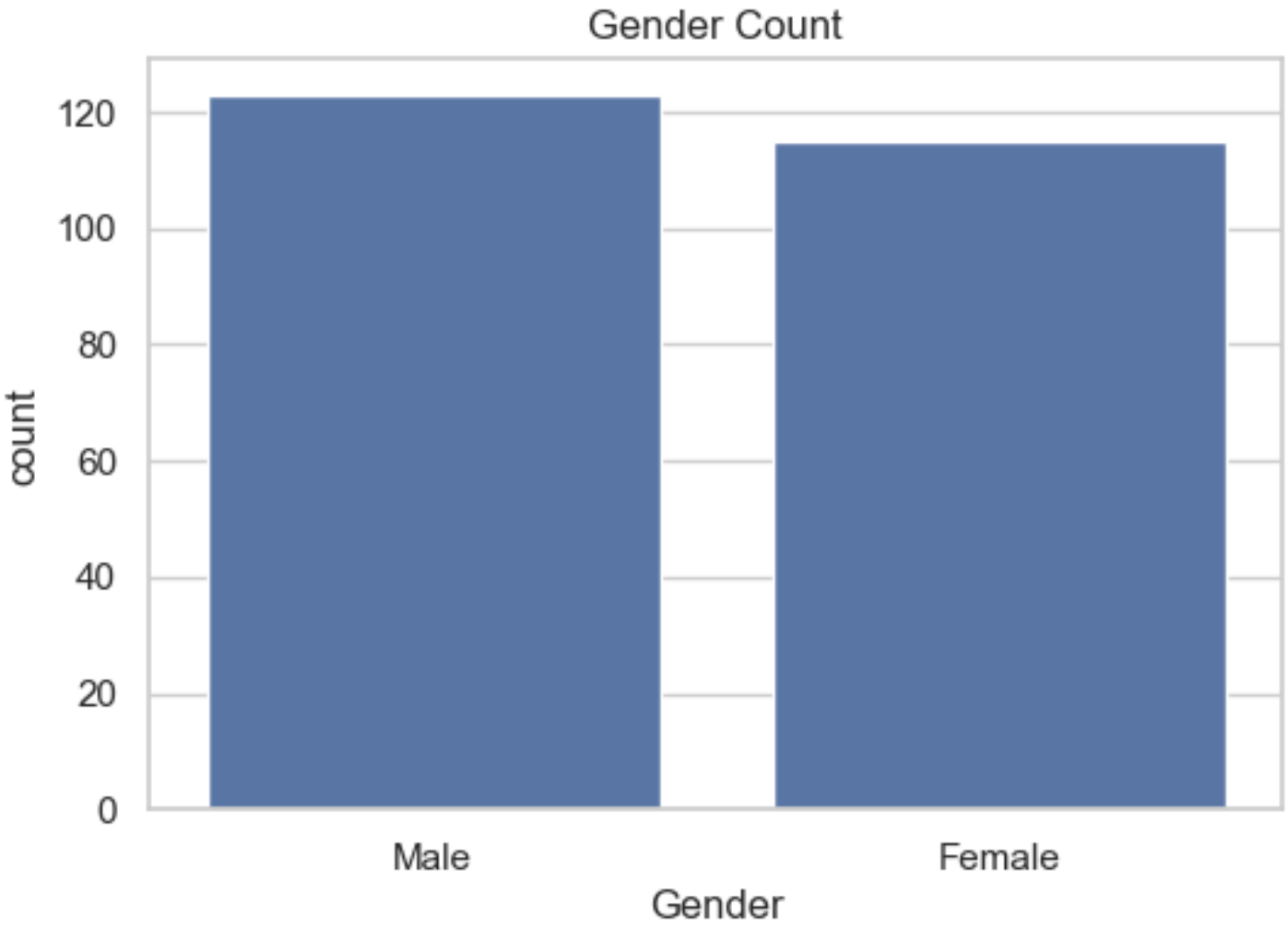
```
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style='whitegrid')

# Univariate: histogram of Age
plt.figure(figsize=(8,4))
sns.histplot(df['Age'], bins=20, kde=True)
plt.title('Age Distribution')
plt.show()

# explanation

# Univariate: bar chart of Gender
plt.figure(figsize=(6,4))
sns.countplot(x='Gender', data=df)
```

```
plt.title('Gender Count')
plt.show()

# Univariate: histogram of AnnualIncome
plt.figure(figsize=(8,4))
sns.histplot(df['AnnualIncome'], bins=20)
plt.title('Annual Income Distribution')
plt.show()

# Bivariate: scatter Age vs. SpendingScore
plt.figure(figsize=(8,6))
sns.scatterplot(x='Age', y='SpendingScore', hue='Gender', data=df)
plt.title('Age vs Spending Score by Gender')
plt.show()

# Bivariate: boxplot Income by Education
plt.figure(figsize=(10,6))
sns.boxplot(x='Education', y='AnnualIncome', data=df)
plt.title('Income by Education Level')
plt.show()
```

```
<Figure size 800x400 with 1 Axes>
```

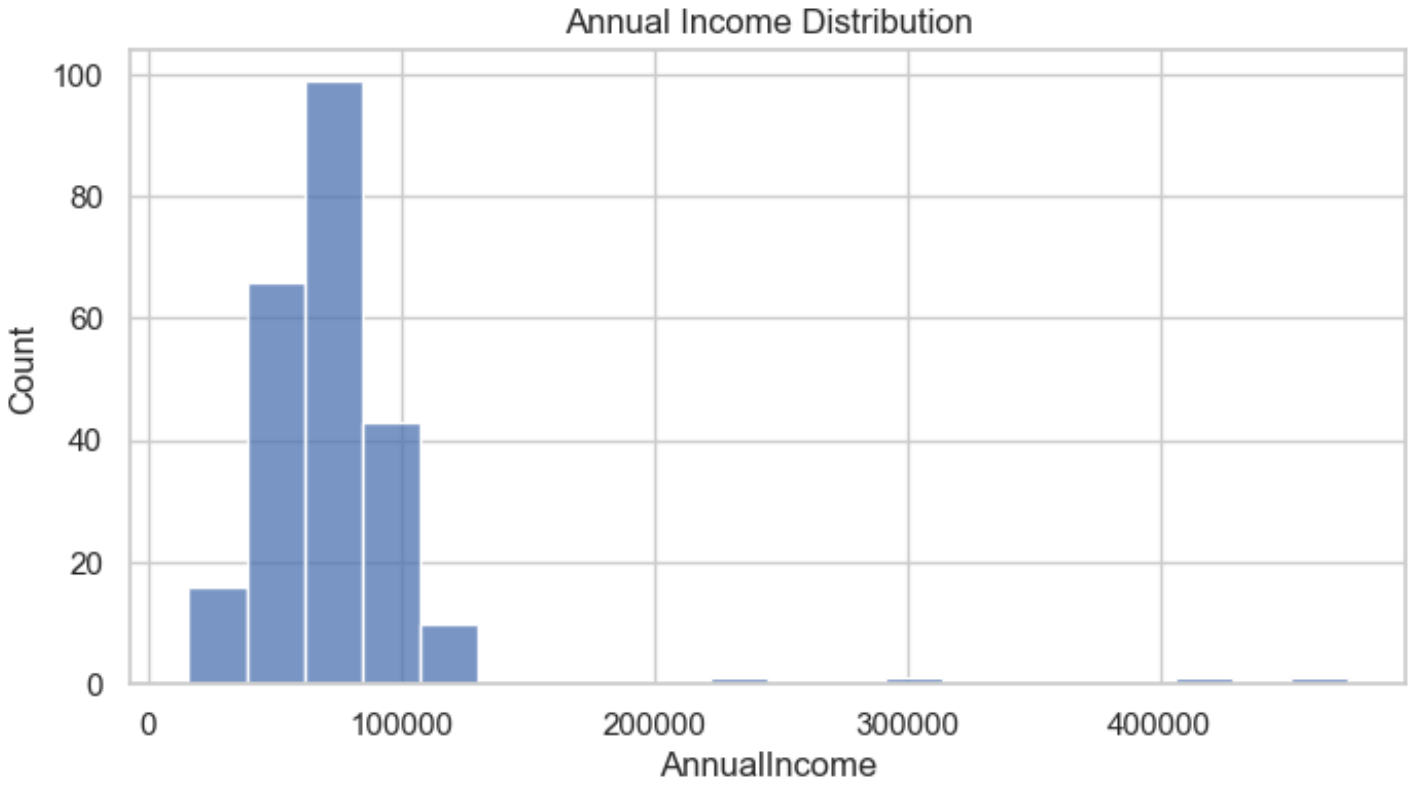



```
<Figure size 600x400 with 1 Axes>
```

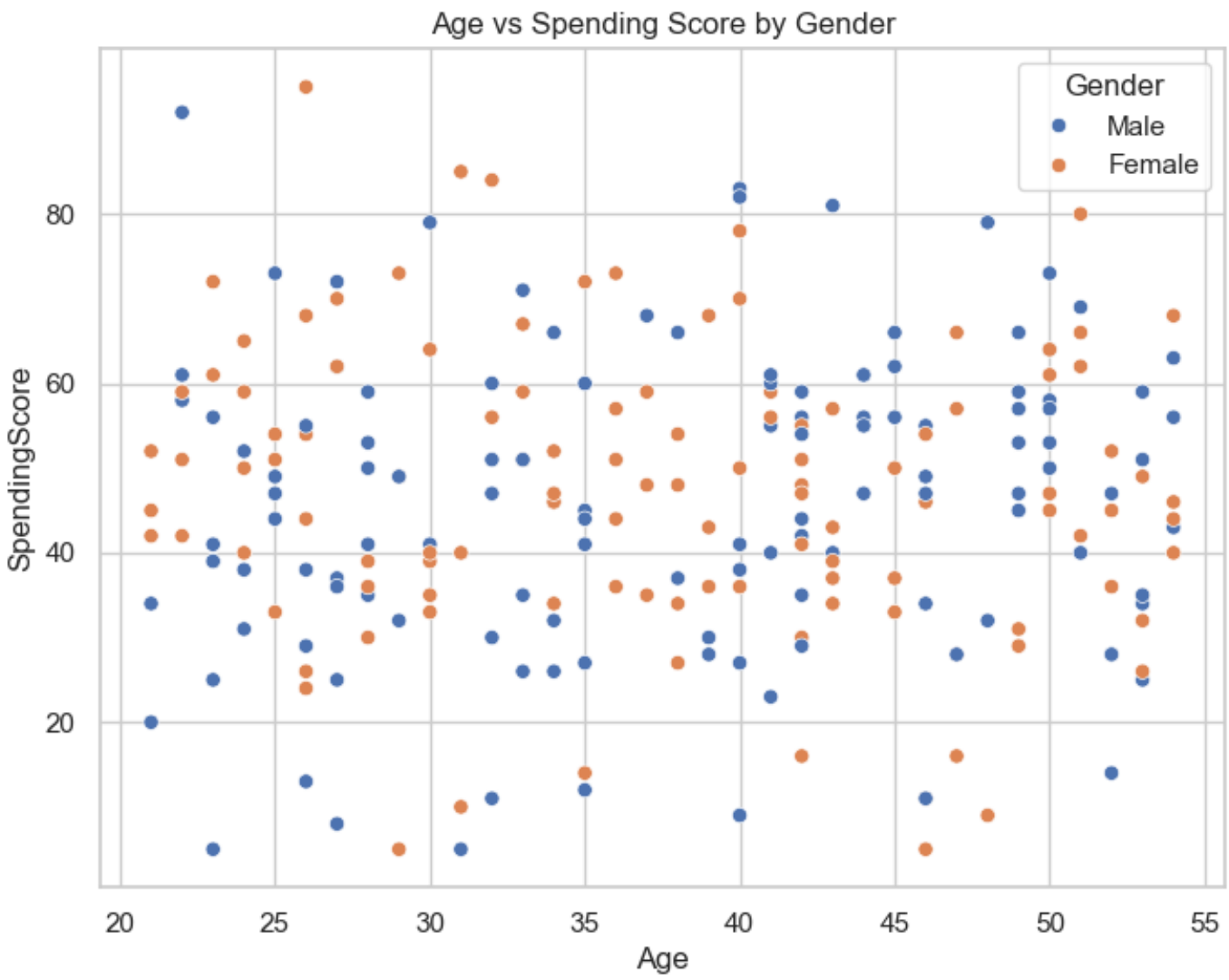



```
<Figure size 800x400 with 1 Axes>
```

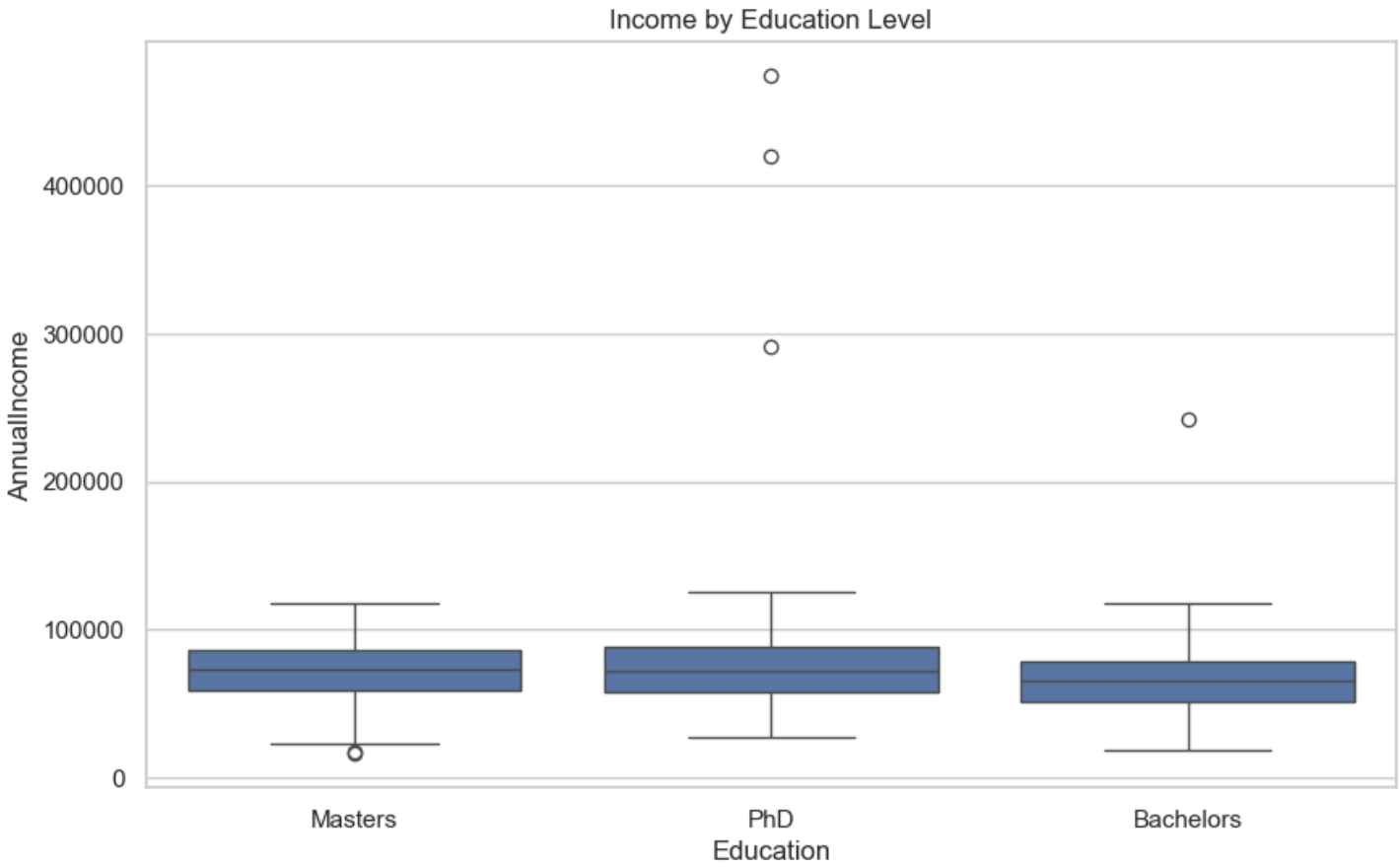
Mini Project EDA Executive Report




```
<Figure size 800x600 with 1 Axes>
```




```
<Figure size 1000x600 with 1 Axes>
```



- **Age Distribution:** The histogram shows most customers are between about 25 and 50 years old with a slight peak in the early 30s, indicating a younger adult customer base.
- **Gender Count:** The bar chart reveals a roughly balanced gender split, though there may be a small male majority.
- **Annual Income Distribution:** Income is right?skewed with many customers earning between 40k and 80k, but a long tail extends to over 100k.
- **Age vs Spending Score:** The scatterplot suggests moderate spending scores across ages; younger customers have higher variability and some of the highest scores, with little difference between genders.
- **Income by Education Level:** The boxplot indicates that customers with PhDs and Masters tend to have higher median incomes compared to those with only Bachelors, and there is considerable income spread within each group.

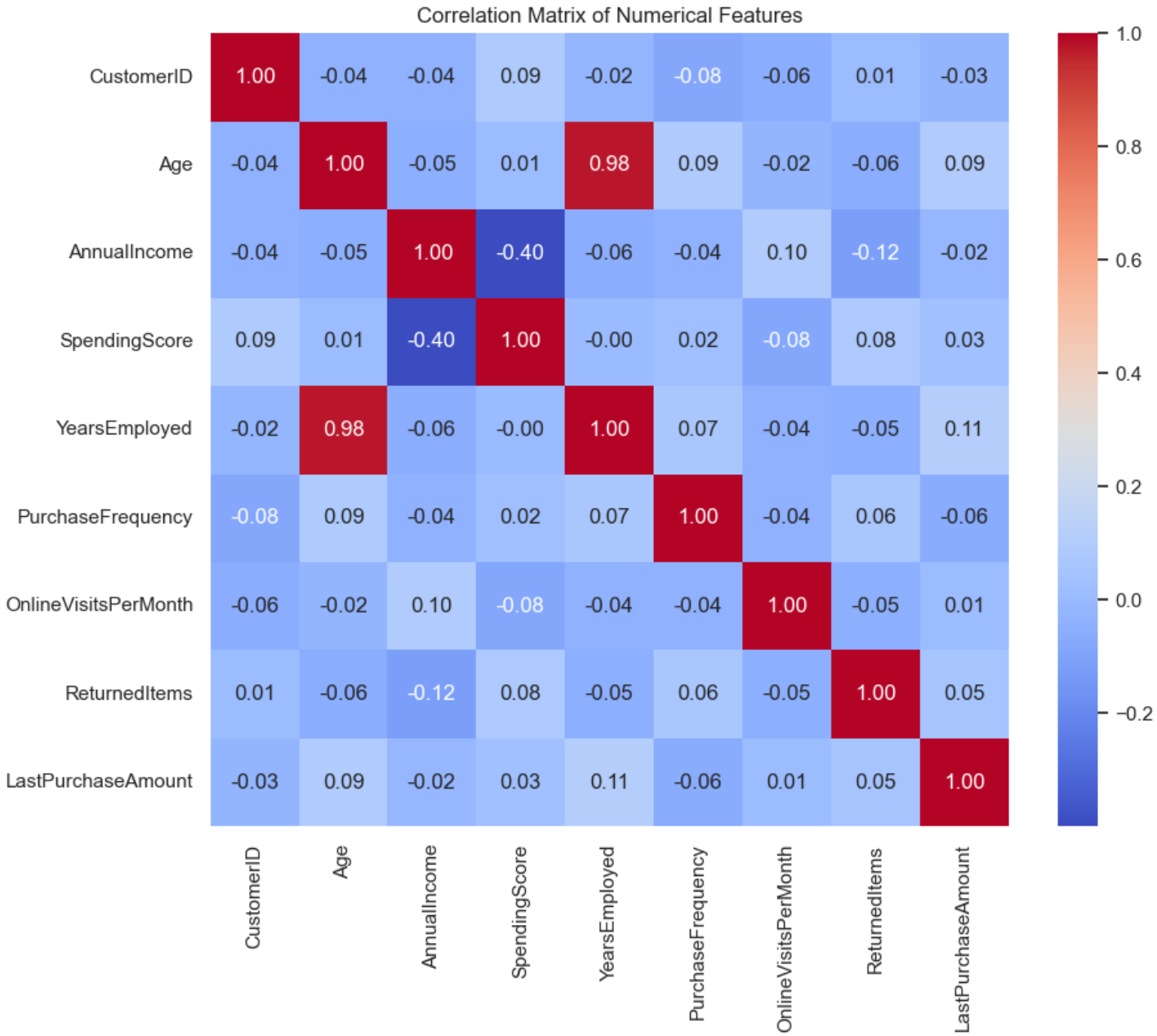
PHASE 4: MULTIVARIATE & STORYTELLING

We compute a correlation matrix for numerical features to identify strong relationships. A heatmap helps visualize these interactions, followed by an executive summary of top findings.

Mini Project EDA Executive Report

```
# compute correlation matrix for numeric columns
numeric_cols = df.select_dtypes(include='number')
cor = numeric_cols.corr()

plt.figure(figsize=(10,8))
sns.heatmap(cor, annot=True, fmt='.2f', cmap='coolwarm')
plt.title('Correlation Matrix of Numerical Features')
plt.show()
```

```
<Figure size 1000x800 with 2 Axes>
```



EXECUTIVE SUMMARY

- **Income and Spending Score:** There is a slight negative correlation between `AnnualIncome`

and `SpendingScore`, suggesting higher-income customers may not necessarily spend more, possibly because they purchase less frequently or more expensive items fewer times.

- **Age Relationships:** Age correlates weakly with several numerical features; younger customers exhibit more variability in spending score and online activity, hinting at diverse engagement levels.
- **Education and Income:** Advanced degrees (Masters/PhD) align with higher annual incomes, reinforcing typical educational return on investment for this customer base.

These insights can guide targeted marketing strategies?e.g., tailoring promotions for younger high-spenders or designing loyalty programs for moderate-income customers with high spending scores.

PROJECT SETUP AND DOCUMENTATION

Additional project artifacts have been created to support reproducibility and sharing:

- `requirements.txt` lists the Python packages needed to run this notebook.
- `README.md` provides an overview of the dataset and instructions for running the analysis.
- `report.md` summarizes key findings outside of the notebook format.

Please ensure the kernel is restarted and all cells are executed in order when re-running this notebook.